

EDUCATION
BACKGROUND**University of Michigan at Ann Arbor**

M.S. in Robotics

Overall GPA: 4.0/4.0;

08/2024 - Present

University of North Carolina at Chapel Hill

B.S. in Applied Mathematics, B.S. in Physics

Overall GPA: 3.918/4.0;

GPA of Applied Mathematics: 3.89/4.0; GPA of Physics: 3.96/4.0

08/2020 - 05/2024

PUBLICATION
*DENOTES EQUAL
CONTRIBUTION

Song, J. *, **Ma, H. ***, Bagoren, O., Sethuraman, A. V., Zhang, Y., Skinner, K. A., “*OceanSim: A GPU-Accelerated Underwater Robot Perception Simulation Framework*,” *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2025. **10/2025 Hangzhou, CN**

Bagoren, O. *, Isaacson, S. *, Sundar, S., Sun, Y., Sheppard, A., **Ma, H.**, Shariff, A., Vasudevan, R., Skinner, K. A., “*SurfSLAM: Sim-to-Real Stereo Underwater Reconstruction For Neural SLAM*,” **In Submission**

CONFERENCE
AND WORKSHOP
PRESENTATIONS“*OceanSim: A GPU-Accelerated Underwater Robot Perception Simulation Framework*”AQ²UASIM Workshop, *IEEE International Conference on Robotics And Automation (ICRA)*,**05/2025 Atlanta, GA**“*Solid Particles Walking on a Vibrating Interface*”Oral, *APS Division of Fluid Dynamics (DFD)*,**11/2024 Salt Lake City, UT**RESEARCH
PROJECT**Training Underwater Object Detection with Synthetic Data****Ann Arbor, MI***Supervised by Prof. Katherine Skinner, Field Robotics Group (FRoG)***08/2025 - Present**

- Built an object-based synthetic data generation pipeline in Nvidia Omniverse, integrated with realistic real-time underwater rendering effects such as haze and particles for sim2real transfer.
- Leveraged SAM+Hunyuan2.1 to replicate 3D assets from image for solving data scarcity and balancing latent distribution in real-world datasets using synthetic data.
- Training object detection model such as YOLO, Mask R-CNN with synthetic data that increases robustness of perception performance in highly degraded and complex-structured environment, such as underwater.
- Extensive applications of OpenUSD for non-destructive and systemic scene authoring and shader coding for domain randomization.

Teleoperating KUKA Arm with Simulation-in-the-loop Digital Twin **Ann Arbor, MI***Supervised by Prof. Dmitry Berenson, Auto Robotic Manipulation Lab (ARM)* **05/2025 - 08/2025**

- Developed simulation-in-the-loop tele-operation of Kuka arm in Isaac Sim environment with Apple Vision VR set, ROS2, and MoveIt motion planning.
- Leveraged Isaac SDF toolkit for high fidelity simulation of forced manipulation task such as closing oil cap or screwing bolts.
- Used PhysX joint simulation for replicating deformable object behaviors such as bending due to gravity and fracture upon intense collision, minimizing sim2real transfer for the trained policy.
- The simulation throughput can reach 20Hz+ for both vision and force signal on a RTX 3090 machine, enabling both downstream policy training and online virtual inference.

Underwater Robots Digital Twin**Ann Arbor, MI***Supervised by Prof. Katherine Skinner, Field Robotics Group (FRoG)***01/2025 - 05/2025**

- 3D Reconstruction of Marine Hydrodynamics Laboratory Towing Tank (MHL, Department of Marine Engineering, UoM) and interactive real-world objects from lidar scan and stereo camera.

- Created customizable digital twin of common underwater environments using Blender and Unreal, and AUV assets such as BlueROV in Isaac Sim, interfaced with ROS2.
- Developed GPU-accelerated underwater sensor (Imaging/Side Scan Sonar, DVL, etc) through Nvidia Warp, a package exposes with python API and JIT compiled to CUDA code.
- Simulated realistic underwater column effects suitable to any Jerlov water types by combining Underwater Image Formation Model and real-time PBR of Omniverse RTX renderer.

Solid Particles Walking on a Vibrating Interface

Chapel Hill, NC

Supervised by Pedro Sáenz, Physical Mathematics Lab (PML)

05/2023 - 05/2024

- Discovered period-doubling vertical dynamics and “walking” states for solid micro-particles $\sim 0.4\text{mm}$ diameter on a vibrated liquid-liquid interface.
- Characterized the “walking” regimes and speed by grayscale image processing and classic object tracking with Kalman Filter.
- Numerically computed and simulated particles’ hydrodynamical behaviors with self-developed PDE numerical solver in MATLAB.
- Designed and assembled the primary experiment structure for vibrational fluid dynamic research with the Modal Shop electrodynamic shaker.

SCHOLARSHIP AND AWARDS

Rackham International Student Fellowship Robotics Department Finalist

- Two finalists per department selected to College of Engineering for fellowship competition. *Department of Robotics, UMich* **08/2025**

Daniel C. Johnson Outstanding Junior Award (Top 1 within department)

- Awarded annually to the physics major who is judged by the faculty to be the most outstanding student of the junior class. *Department of Physics and Astronomy, UNC* **04/2023**

Summer Undergraduate Research Fellowship (54 recipients, 2023)

- Granted to undergraduate students conducting research under a faculty mentor. \$4000 fellowship included. *Department of Undergraduate Research, UNC* **04/2023**

SKILLS

Software/Computing Literacy: Isaac Sim/Lab, OpenUSD, Omniverse Kit, ROS2, Python, C/C++, PyTorch, GLSL, CUDA, PhysX, Docker, SolidWorks, Arduino, LabView, Unreal, Unity, Blender, Adobe Illustrator, Premiere Pro, Substance Combo

Experimental Techniques: High-Speed Videography, 3D Printing, Laser Cutting, Basic Electronics, Metal and Wood Crafting, NI-DAQ based Signal Processing, Particle Image Velocimetry

Professional Membership: *IEEE* and *APS* (American Physical Society)

Services: Reviewer of *IROS*, *ICRA*

REFERENCES

Field Robotics Group (FRoG)

Prof. Katherine Skinner

Assistant Professor

Robotics, UoM

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